

Interactive generation and stereoscopic observation of 3DCG scenes based on 2D tabletop drawing

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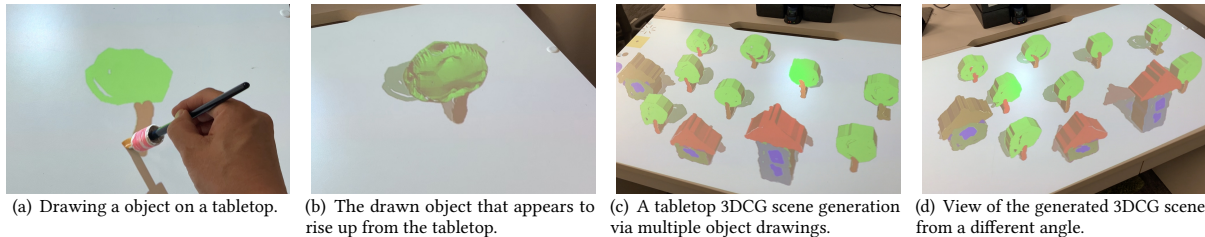


Figure 1: Generating 3DCG scenes based on tabletop 2D drawing using the proposed system.

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1 Introduction

Digital tools for drawing enable a wide variety of expressive techniques in drawing through the use of computer graphics CG technologies, and various interactive methods such as generating 3DCG scenes from hand-drawn 2D sketches[Mizuno and Funahashi 2017], and AR technology to display 3DCG models generated from 2D sketches as if they appear in the real space[whatever 2019]. However, in existing digital tools, the creation of 3D models through drawing and the observation of the generated models are often separated, or there exist barriers between these two processes. Users observed the 3DCG models generated from their drawings either displayed on a screen or appearing in the real space through a smartphone display.

We propose a novel drawing tool that seamlessly integrates the generation and observation of 3DCG scenes from 2D sketches. When a user draws on a tabletop, the depicted object instantly appears standing on it. By continuing to draw, the user can interactively build a 3DCG scene viewable as a stereoscopic image.

CG generation based on drawing is very popular in entertainment facilities; however, as it is becoming more common, even children are no longer easily surprised by it. In contrast, an interactive system that seamlessly brings a 3DCG scene onto a tabletop

based on drawing, as in this study, has not been reported before. The motivation of this study is to develop a CG generation interface that can provide unprecedented surprise and capture interest in entertainment facilities.

2 Developed System and Method

In this system, a projector displays a drawing canvas and interface elements for adjusting pen color and thickness onto a tabletop surface. Users can freely draw on the canvas using a digital pen or similar input device (Fig. 1(a)). Once the user finishes drawing an object, the drawn object appears to rise up from the table (Fig. 1(b)). The raised object is presented as a stereoscopic image, and it remains visually consistent as a 3D object standing on the table, even when the user's viewpoint changes, giving the impression that it physically exists in the real space. As more objects are drawn, additional standing objects emerge on the table, allowing the user to interactively construct a 3DCG scene (Fig. 1(c)(d)).

To realize the proposed system, it is necessary to integrate three key components: a touch-enabled tabletop drawing interface, generation of a 3DCG scene based on 2D object drawings, and stereoscopic visualization of the 3DCG scene on the tabletop.

The touch-enabled tabletop drawing interface is realized by projecting images onto the tabletop using a projector and scanning the tabletop with a 2D LiDAR. When a user draws within the canvas area on the tabletop using a pen-shaped device such as a conventional brush, the 2D LiDAR is used to acquire the pen tip coordinates. The pen's movement trajectory is then visualized as a polyline composed of circles and rectangles. This allows the user to draw 2D objects on the tabletop. The 2D LiDAR is also used to detect touch inputs on projected interface elements. When the user finishes a drawing operation and lifts the pen-shaped device from the tabletop, the system waits for a fixed period of time. If no further input is detected during this interval, the system determines that a single object has been completed.

Once an object is completed, the system determines a bounding rectangle around it and extracts the corresponding region from the

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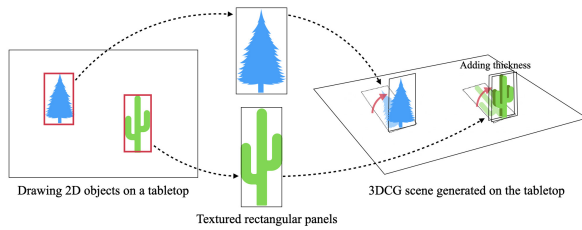


Figure 2: A 3DCG scene based on 2D object drawings.

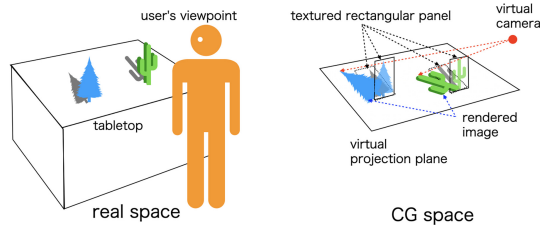


Figure 3: Reproducing the real-world situation of the table in the CG space.

canvas to create an image. The background is made transparent, and the image is applied as a texture onto a rectangular panel of the same size. By aligning this textured panel with the original region and raising it vertically, the object appears to stand upright on the tabletop. A 3D model can be generated by adding thickness to the textured panel, which can also vary by location based on the distance transform of the drawn region (Fig. 1(b)). Each time the user draws with the pen, the system converts the object into a textured panel, and by successively placing these upright, a 3DCG scene is incrementally constructed (Fig. 2).

Stereoscopic observation of the generated 3DCG scene on the tabletop is achieved through a combination of motion parallax-based stereoscopy and spatial consistency based on anamorphosis, a form of optical illusion art. To achieve motion parallax-based stereoscopic visualization, the system uses a 3D camera (Kinect) to track the user's viewpoint and renders the 3DCG scene in real time according to the user's viewing position. Then, in order to display an image with spatial consistency based on anamorphosis on the tabletop, the CG space reproduces the real-world situation in which the user observes the drawn objects standing on the table (Fig. 3). Specifically, the system reconstructs a 3D scene in the CG space by arranging textured rectangular panels or 3D models to represent the drawn objects standing on the table. A virtual camera is placed at a position corresponding to the user's viewpoint, and a virtual projection plane is placed to match the physical tabletop. The textured rectangular panels or 3D models are then rendered from this virtual camera. When the resulting image is projected onto the real tabletop, it appears spatially consistent from the user's perspective, achieving an anamorphic illusion. To enhance the sense of depth, a desaturated and darkened version of each textured rectangular panel is placed at the same position as the user-drawn object to represent its shadow. Stereoscopic images based on motion parallax and anamorphosis-based spatial consistency can be observed with the naked eye, and particularly when viewed with one eye, a strong sense of depth is perceived.

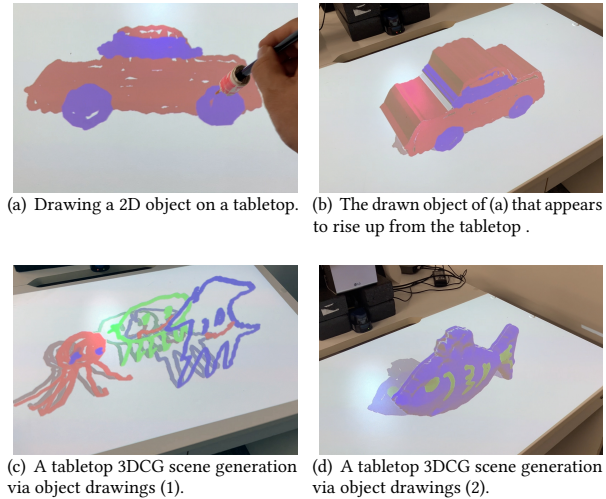


Figure 4: Examples of generating 3DCG scenes based on tabletop 2D drawing using the proposed system.

3 Experiments and Result

We implemented the system and conducted experiments. The system was implemented in C++ on a Mac mini (Intel Core i7) using OpenGL and OpenCV. A Kinect V1 was used as the 3D camera, and a 2D LiDAR sensor (UBG-04LX-F01, Hokuyo) was used for tracking. An ultra-short-throw projector was employed to project images onto the tabletop with a display area of 124×70(cm). Ordinary sticks and brushes were used as pen-like input devices.

We confirmed that the proposed system enables seamless integration of drawing, 3DCG scene generation, and stereoscopic observation, giving users the sense of building a 3DCG scene on the table through drawing (Fig. 4). Each time a drawing action was completed, the drawn object appeared to rise from the surface of the table, allowing users to perceive the object as if it physically existed on the tabletop.

The prototype system was evaluated by approximately 30 users and received highly positive feedback, including remarks such as “a novel experience” and “children would be delighted by this”. Therefore, we believe that the proposed system can be directly applied to entertainment purposes. Furthermore, by sharing the image data that serves as the basis for the 3DCG scene over a network, remote users can also create and share tabletop 3DCG scenes through drawing. Thus, the proposed system has the potential to be utilized for collaborative tasks such as design.

In this study, a simplified method was used to generate 3D models from drawings, primarily for the purpose of verifying entertainment applications. In the future, more precise 3D model generation using AI is planned, allowing application across a wider range of fields.

References

- S. Mizuno and K. Funahashi. 2017. Amazing Sketchbook Advance. *SIGGRAPH 2017 Posters* Article No. 2 (2017).
- whatever. 2019. Rakugaki AR. <https://whatever.co/work/rakugakiar/> (2019).